

Mehdi Eskandari-Ghadi

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www.linkedin.com/in/M-E-Ghadi | Authorized to work in the U.S. | Sponsor not required

EDUCATION

Civil Engineering Ph.D. (GPA 4.0)

Engineering Science Focus
University of Colorado Boulder

Boulder, CO

2016 – 2021

TECHNICAL SKILLS

Programming	Python, TypeScript/JavaScript, VBA, MATLAB, AutoLISP
Platforms & Data	Palantir Foundry, Apache Spark, Polars
Engineering	AutoCAD, PLSCADD, L-PILE, ABAQUS/LS-DYNA
Standards & Tools	NESC, GO-95, FERC-881, Git, Microsoft Office Suite, LaTeX

ENGINEERING EXPERIENCE

Reliability & Standards Software Engineer II

Jan 2025 – Present

PacifiCorp

- Built full-stack engineering workflows on the Palantir Foundry platform using TypeScript and Python, covering audit preparation, data management UIs, interdepartmental data collection, and automated queue-based processing pipelines.
- Designed a git-inspired facility lifecycle management system within Foundry for branching, modifying, and publishing transmission facilities, with automated resource tracking, branch conflict resolution, and staged approval workflows.
- Architected backend data pipelines in Python using Spark and Polars for large-scale ingestion, parsing, and health remediation of legacy spreadsheets, automating correction workflows and multi-source data merging to bring records to quality standards.
- Implemented IEEE-compliant thermal rating models in Python covering solar heat gain, convective cooling, radiative heat loss, and conductor ampacity-resistance energy balance, enabling dynamic and static line rating computations within the platform.
- Implemented Python and TypeScript routines within Foundry to perform geometric analysis of parallel transmission corridors, computing mutual impedance candidate pairs and characterizing average and percentile conductor separation distances, with line-to-line distance profiles visualized in platform-integrated UI graphs.
- Overhauled legacy spreadsheets with VBA-based functionalities, maintainable assumption/reference sheets, and improved UX, establishing a more auditable and automatable calculation baseline ahead of platform migration.
- Developed AutoLISP routines for AutoCAD one-line drawings that parse shape geometries and layer data to identify transmission lines, facilities, and equipment; deduce line segments and breaker-to-breaker sections including different equipment configurations; and export structured data to spreadsheets to bring drawing databases and engineering records to convergence.

Transmission Engineer I/II

Jan 2023 – Jan 2025

PacifiCorp

- Utilizing high-level knowledge of transmission line design to scope new transmission lines in collaboration with other engineering disciplines.
- Utilizing in-depth knowledge of PLSCADD, L-PILE, NESC, and GO-95 for design of new power transmission lines and rebuilding/maintenance of existing lines in states of UT, WY, OR, WA, and CA in collaboration with teams of engineers, managers, field crew, and legal.
- Utilizing in-depth knowledge of structural engineering and geotechnical engineering for custom-engineered solutions of transmission structures and drilled pier foundations with reference to ASCE, ACI, and FHWA to meet special needs.
- Reduced engineering hours by threefold and cost by approx. \$1.5M by intersecting programming and automation experience and transmission engineering experience to create a line rating analysis tool. This analysis tool allows compliance with FERC-881 with minimum reduction of line capacity, increased calculation accuracy, and reduced engineering time.
- Utilized visual basic programming skills to create automated tools for data analysis and modeling of existing transmission line ratings to assess FERC-881 compliance.
- Applying programming and software design skills and IEEE line rating knowledge to guide the development of proprietary software tools for FAC-008 compliance.
- Employed excel programming and data analysis skills to develop automated tools for deep foundation design, construction package generation, and transmission pole design aids.
- Utilized transmission engineering knowledge and programming experience to generate company sag-tension standards for tree-wire.
- Designing and reviewing drawings for tubular steel poles and drilled piers to withstand vertical and lateral loads.

- Performing detail-oriented technical review of transmission standards and collaboration in their developments.
- Initiating process improvement pipelines for transmission design processes and documentations.
- Improved company standards and company procedures with detail-oriented reviews/QAQC.

RESEARCH EXPERIENCE

Academic Research

Graduate Research Assistant / Ph.D. Candidate

Jun 2018 – Jan 2023

University of Colorado Boulder, Boulder, CO

- Collaborated with a team of three senior researchers at the Lawrence Berkeley National Laboratory to investigate and develop mechanistic models for subcritical crack growth healing.
- Utilized knowledge of fundamentals of fracture mechanics, fluid transport, and boundary element methods to develop a software that couples fields of mechanics and physics.
- Merged expertise in research, programming, and engineering to investigate multiphysical and multiscale solid-fluid interactions (e.g., adsorption, surface tension, solvation pressure) in porous solids and develop a thermodynamics-poromechanics-based MATLAB software for “adsorption-induced deformation of microporous material”.
- Employed technical writing and illustrative skills for preparation of 4 peer-reviewed publications in journals of JMPS, IJSS, and EFM.

Talks and Presentations

EMI Virtual Conference 2021 and EMI Conference 2022

May 2021 and May 2022

- Applied illustrative and verbal communication skills to deliver presentations titled “A multi-scale theory explaining the initial shrinkage of micro-porous solid upon gas adsorption” and “A Multiphysical Surface-Force Based Fracture Theory for Subcritical Crack Growth in Surface-Reactive Environments”, followed by Q&A from engineering and academic audience.
- Received runner up award in the 2021 Student Poster Competition for poster presentation on sorption-induced deformation of porous material.

Lawrence Berkeley National Laboratory BES

Sep 2020 – Jan 2023

- Presented regular scientific presentations on topic of crack propagation to teams of research professionals and senior scientists in the LBNL BES with diverse backgrounds in mechanics, physics, and chemistry.

Extracurricular Projects

2018

- Programmed a discrete element MATLAB code for modeling of grain structure stability and avalanche.

TEACHING EXPERIENCE

Teaching Assistantships/Tutoring

Teaching Assistant / Tutor

Aug 2016 – Dec 2019

University of Colorado Boulder, Boulder, CO

- Presented class lectures and laboratory teaching sessions, held hours of office hours and exam preparation sessions, prepared and graded exams and homework question sets, for courses of Advanced mechanics of materials (graduate level) and Dynamics (undergraduate level).
- Tutored students for FE exam with 100% success rate.
- Tailored technical communication specific to each student’s understanding and utilized CAD skills to enhance visualization and understanding.

Teaching Assistant

Fall 2014 and Fall 2015

University of Tehran, Tehran, Iran

- Held weekly office hours and graded homework, quizzes, and class projects, for courses of Structural Analysis II (undergraduate level) and Hydraulics of open channels (undergraduate level).

LEADERSHIP ROLES

Mentorship

Research Mentor

Jan 2022 – Jan 2023

University of Colorado Boulder, Boulder, CO

- Trained new Ph.D. students on fundamentals of mechanics, written and oral skills, and technical research.
- Guided a research trainee through the modeling and simulation of beam bending in humidity actuators.

President/Co-president — Persian Student Organization, Boulder, CO

Aug 2018 – May 2019

University of Colorado Boulder, Boulder, CO

- Proposed, secured funding, planned, and oversaw student events in collaboration with professionals from industry and academia, local businesses, and non-profit organizations.

OTHER INTERESTS

- I am a plant hobbyist, animal lover, dog owner, and skilled chef for humans and dogs.